A-level

# ENVIRONMENTAL SCIENCE

## Paper 2 - Generated practice paper Set 25

## Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance | Marks |
|---|---|---|
| **01.1** | **Calculation**<br>A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark.<br>**1** Σn(n - 1) = (40 x 39) + (30 x 29) + (20 x 19) + (10 x 9) = 1560 + 870 + 380 + 90 = 2900<br>**1** N(N - 1) = 100 x 99 = 9900<br>**1** D = 9900 / 2900 = 3.41 (accept 3.4)<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• **Simpson** → a diversity index that uses both the number of species and their abundance<br>**Do not credit**<br>• A bare wrong answer with no working.<br>• 'more species always means a higher index' - the evenness of the abundances matters as well | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **01.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B<br>**1** The index does not identify which species are present / a rare or important species may be missed<br>– depends on identification skill and sampling method<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• The converse of any point, ie the same comparison written the other way round.<br>• **biodiversity** → the variety of life: habitat, species and genetic diversity | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3. | |
| **02.1** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1** Sinkers / weighted lines so hooks sink quickly below the reach of albatrosses<br>**1** Setting lines at night when birds are not feeding<br>**1** Bird-scaring streamers / decoys<br>**1** Curved hooks that catch tuna but not birds<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. | 3 |
| **02.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1 Effort:** limits on boat size, engine power, days at sea, number of licences<br>– decommissioning payments to reduce fleet size where overfishing is unavoidable<br>**1 Methods:** bans or length limits on drift nets<br>– bans on demersal trawling over sensitive seabed (Rockall Bank, Hatton Bank deep-water reefs)<br>**1** Shorter trawl tows so turtles / by-catch survive (over 99% survive under 10 minutes)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **decommissioning** → safely dismantling a power station at the end of its life<br>**Do not credit**<br>• quotas described as solving overfishing on their own - undersized fish are still caught and discarded | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **02.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Eggs / larvae are reared in captivity through the planktonic stage when most die (predation, failure to find habitat)<br>**1** Juveniles are released when survival chances are good, increasing recruitment to the adult population<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **02.4** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Egg / larval survival varies greatly between years so adult numbers do not predict recruitment<br>**1** Several poor years would lower the MSY<br>– egg surveys allow catches to be reduced in advance rather than after a sudden collapse<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **MSY** → the largest catch that can be taken without reducing the stock | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2. | |
| **03.1** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1 Ecological footprint:** the area of the Earth's surface (global hectares) needed to provide the resources used by a person / population / activity and to absorb the wastes produced<br>**1 Biocapacity:** the biological productivity of an area (forests, cropland, grazing, fisheries, built land) - its ability to supply resources and absorb wastes, expressed in global hectares<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **03.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** A hectare of land / sea with the world-average biological productivity<br>**1** Biocapacities of real areas vary widely so a standard unit allows footprints and biocapacities of different places to be compared<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **03.3** | **Calculation**<br>1 mark for the correct answer; working does not have to be shown.<br>**1** 8.1 / 1.7 = 4.8 (accept 4.7-4.8)<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (Earths) is already printed on the answer line.<br>**Do not credit**<br>• A bare wrong answer with no working. | 1 |
| **03.4** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Higher HDI is associated with a larger footprint (USA 8.1 vs Bangladesh 0.8), though the UK achieves the same HDI as the USA with about half the footprint<br>**1** Global footprint (2.8) exceeds global biocapacity (1.7) so humanity uses resources faster than they regenerate and CO2 accumulates (overshoot)<br>**1** Earth Overshoot Day is the date each year when a year's worth of resources / waste absorption has been used up<br>– it falls earlier each year (8 August in 2016)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |
| **03.5** | **Any 2 of 4**<br>1 mark each for any 2 of the 4 points below, in any order, maximum 2.<br>**1** Renewable energy and energy conservation (carbon footprint is the largest component)<br>**1** Reduced meat consumption / food waste<br>– local seasonal food<br>**1 Circular economy:** repair, re-use, recycling<br>– longer product lives<br>**1** Public transport, efficient buildings<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **energy conservation** → using less energy, eg by insulation or behaviour change<br>• **recycling** → reprocessing a material to make something new<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.<br>• 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts<br>• 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 379-380; AQA spec 3.6.4.2. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **04.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** The nucleus of a stem cell from the endangered species is transferred into an empty egg of a closely related species (southern white rhino)<br>**1** The embryo is implanted into a surrogate female of the related species which gives birth to a northern white rhino<br>**1** Very low success rates with wildlife so far<br>– deep-frozen cells could allow clones even after the last individuals die<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |
| **04.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1 Micro-propagation:** tissue culture producing many cell clusters from one plant, each grown into a genetically identical plant<br>**1 Black poplar:** cuttings / pollarded stems of pure native trees are rooted to produce many identical trees<br>**1** Needed because the flowers are pollinated by introduced subspecies producing hybrids, so cloning maintains the pure gene pool (but no genetic diversity)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **gene pool** → the range of alleles / genes present in a population | 3 |
| **04.3** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** Stores seeds of wild plants (especially threatened species and crop wild relatives) so species are not lost if extinct in the wild<br>**1** Seeds collected worldwide<br>– target of 10 000 seeds per species to maintain a diverse gene pool<br>**1** Stored dry and refrigerated / frozen in an underground reinforced vault to reduce risk of accident<br>**1** Seeds germinated periodically to check viability and replenish stocks<br>– can be used for reintroduction / breeding<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **gene pool** → the range of alleles / genes present in a population | 4 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 35-37; AQA spec 3.1.2.3.3. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **05.1** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1 General physiological solvent:** most biological reactions take place between substances dissolved in water<br>**1** Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes<br>**1** High specific heat capacity moderates temperature changes<br>– evaporation absorbs heat (cooling)<br>**1 Anomalous expansion:** ice floats so water bodies do not freeze solid<br>– absorbs UV protecting early marine life<br>– provides aquatic habitats<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **Anomalous** → an outlier / a result that does not fit the pattern<br>**Do not credit**<br>• Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.<br>• discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty | 4 |
| **05.2** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen<br>– at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago)<br>**1** Oxygen in the stratosphere absorbed UV, forming ozone (O3) in a dynamic equilibrium of O, O2 and O3<br>**1** The ozone layer reduced UV reaching the surface so organisms could survive out of the water<br>**1** Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved<br>– many anaerobic archaea died out<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **ozone layer** → the ozone in the stratosphere that absorbs UV<br>**Do not credit**<br>• answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated | 4 |
| **05.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Photosynthesis removed CO2 (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels<br>**1** This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **infrared** → heat radiation / long-wave radiation re-emitted by the Earth | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2. | |
| **06.1** | 1 mark for the point below.<br>**1** Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **sustainable** → meeting present needs without stopping future generations meeting theirs<br>**Do not credit**<br>• 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future | 1 |
| **06.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Impact depends on total population x per capita impact<br>**1** With few people, natural processes (forest regrowth, fish recruitment, aerobic decomposition / dilution) recover faster than the resource is used or the waste added<br>**1** Larger populations exceed the rate of natural recovery: forest and soil do not regenerate, catches exceed MSY, water deoxygenates<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **sustainable** → meeting present needs without stopping future generations meeting theirs<br>• **MSY** → the largest catch that can be taken without reducing the stock<br>**Do not credit**<br>• 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future<br>• quotas described as solving overfishing on their own - undersized fish are still caught and discarded | 3 |
| **06.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Urban areas are supported by resources brought in from elsewhere<br>– when one area is depleted, exploitation simply moves (timber, fish, minerals, energy) so consumers see no shortage<br>**1** Rural communities exploit local resources directly so the effects of over-exploitation become obvious quickly, encouraging cultural limits to need<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **06.4** | **Marks shared**<br>Up to 4 marks in total - share them between the 3 points below according to how completely each one is made.<br>**up to 2 Climate:** warming increases evaporation, cloud and albedo<br>– higher CO2 increases photosynthesis and sequestration<br>– hydrological cycle: more evaporation gives more precipitation<br>– populations: density-dependent factors return numbers to carrying capacity<br>**up to 2** In agriculture soil degradation, water shortage and loss of predators would reduce yields (negative feedback limiting the activity)<br>**up to 2** Humans respond with fertilisers, irrigation and pesticides to maintain yields, further damaging the natural systems that support production<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **negative feedback** → a change that cancels itself out / a self-correcting cycle<br>• **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back<br>• **carrying capacity** → the largest population the area can support<br>**Do not credit**<br>• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on<br>• 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed | 4 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1. | |
| **07.1** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1 Isolation makes colonisation difficult:** only marine species, seabirds, plants with floating seeds or chance arrivals (tortoises on logs, wind-blown birds) reach them<br>**1** Whole taxa (terrestrial mammals, amphibians) may be absent so few original colonists<br>**1** Colonists evolve in isolation under different selection pressures to fill the empty niches, producing unique species (Galapagos, Madagascar lemurs, Hawaii)<br>**1** Endemics have small ranges and populations so are vulnerable to extinction<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **endemic** → found naturally only in that one place<br>• **biodiversity** → the variety of life: habitat, species and genetic diversity | 4 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **07.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** The birds evolved without mammal predators so nest on the ground / in burrows with no defence<br>**1** Rats eat eggs and chicks<br>– numbers fell to very low levels (45 and 12 pairs)<br>**1** After eradication breeding success rose about 7-12 fold, confirming predation was the main limiting factor<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **limiting factor** → the factor in shortest supply that holds the rate or population back | 3 |
| **07.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Introduced mosquitoes carry avian malaria to which endemic birds have no resistance<br>– birds survive only above about 1500 m where it is too cold for malaria<br>**1** Warming allows mosquitoes / malaria to spread to higher altitudes, shrinking the safe range<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **endemic** → found naturally only in that one place | 2 |
| **07.4** | 1 mark for the point below.<br>**1** Control of developments and visitor numbers (biosecurity, quarantine of imports)<br>– protected areas<br>– captive breeding of endemics<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **captive breeding** → breeding in captivity to increase numbers for release<br>**Do not credit**<br>• captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 64-65; AQA spec 3.1.2.3.4. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **08.1** | **Marks shared**<br>Up to 4 marks in total - share them between the 3 points below according to how completely each one is made.<br>**up to 2 Extensive:** low inputs of energy, food and chemicals so low cost and low environmental impact<br>– sustainable<br>– but low yield per area and dependent on natural productivity<br>**up to 2 Intensive:** control of all limiting factors gives very high yield and profit per area<br>– but high inputs (fishmeal from wild fish, energy), disease and lice at high density, pollution from wastes, antibiotic / pesticide use, escapes<br>**up to 2** Judgement depends on species, market and site<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **aquaculture** → fish or shellfish farming<br>• **sustainable** → meeting present needs without stopping future generations meeting theirs<br>**Do not credit**<br>• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on<br>• 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future | 4 |
| **08.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Adults with desirable characteristics (fast growth, shape, colour) stripped of eggs and milt which are mixed for fertilisation<br>**1** Eggs raised in aerated freshwater tanks, dead / diseased eggs removed daily<br>– fry fed fishmeal pellets and moved through tanks as they grow<br>**1** After 12-18 months smolts transferred to seawater cages, fed pellets until harvest at 3-5 kg<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |
| **08.3** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** About 10% of the world's mangroves have been cleared for coastal shrimp lagoons (up to 100 ha), removing nurseries and coastal protection<br>**1** Lagoons need water exchange to remove wastes and aeration to prevent deoxygenation as wastes decompose<br>**1** Eyestalk ablation stimulates breeding<br>– high densities increase disease so antibiotics / pesticides used<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **deoxygenation** → loss of dissolved oxygen from the water<br>**Do not credit**<br>• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on | 3 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-356; AQA spec 3.5.2.3. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **09.1** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Island communities evolved from few colonising species in isolation, often without mammal predators or large herbivores<br>**1** So indigenous species lack adaptations (eg ground-nesting birds, flightlessness) to survive introduced cats, rats, pigs, goats<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **09.2** | **2 marks per point**<br>2 marks for each of the 3 points below; award 1 mark where a point is only partly made.<br>**2 Competitor:** grey squirrel (North America) out-competes red squirrel - digests acorns, larger, wins nest sites<br>– rhododendron shades ground and releases toxins preventing native regrowth<br>**2 Predator:** American mink (escaped fur farms) reduced water vole range<br>– cane toads in Australia<br>– cats / rats / pigs on islands eating ground-nesting birds<br>– Nile perch in Lake Victoria ate cichlids to extinction<br>**2 Pathogen:** squirrel pox virus carried by grey squirrels kills red squirrels<br>– crayfish plague carried by signal crayfish kills white-clawed crayfish<br>– Dutch elm disease, ash dieback, sudden oak death imported on plants / soil<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 6 |
| **09.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Closely related introduced species (sika deer; domestic cats) cross-breed producing fertile hybrids<br>**1** The natural gene pool is changed / diluted with genes it would not naturally contain, so the pure indigenous species may disappear<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **gene pool** → the range of alleles / genes present in a population | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 24-25; AQA spec 3.1.2.2. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **10.1** | **Levels of response**<br>Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist.<br>**Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.<br>**Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.<br>**Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.<br>**0 marks**: No relevant content.<br>**Indicative content - a guide to placing the mark, not a checklist**<br>• **Succession:** pioneers, seres (hydrosere, lithosere, psammosere), modification of abiotic conditions, climax communities controlled by climate<br>– deflected succession and plagioclimaxes<br>• **Application:** deciding whether to manage (plagioclimax habitats need grazing, mowing, burning, coppicing, pollarding continued) or leave alone (climax habitats)<br>– habitat creation planning the stages<br>– controlling scrub / secondary succession<br>– understanding that new habitats change over time<br>• **Population dynamics:** birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles, r- and k-selection, MSY<br>• **Application:** setting sustainable harvest levels (fisheries, whaling, forestry)<br>– recognising k-selected species need strict protection and recover slowly<br>– deciding when culling is needed (deer without wolves, introduced species) or when releases / captive breeding are needed (low breeding rate)<br>– predicting effects of reintroducing predators<br>• Monitoring population size, distribution, survival rate and age structure to assess success<br>– adaptation and range of tolerance guiding abiotic management<br>• **Evaluation:** uncertainties (natural fluctuations, unknown carrying capacity)<br>– examples (Yellowstone wolves, Scottish deer, chalk grassland)<br>– judgement<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• Any accurate, relevant material that is not in the indicative content. | 9 |
| | • **succession** → the gradual change in the community over time<br>• **abiotic** → non-living / physical factors<br>• **deflected succession** → a community held back from its climax by human activity<br>**Do not credit**<br>• 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future<br>• captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | **Model answer - Level 3 (8 marks)** | |
| | Succession is the sequence by which pioneers colonise bare ground, modify the abiotic conditions and are replaced by later seres until a climax community controlled by climate is reached; where grazing, mowing, burning or coppicing halts it, a plagioclimax forms. This tells a conservation manager whether to intervene: chalk grassland, heathland and coppiced woodland are plagioclimaxes that disappear without continued management, whereas a climax habitat is best left alone, and habitat creation must plan the stages, control scrub and manage secondary succession. | |
| | Population dynamics - birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles and r- and k-selection - inform decisions of a different kind. Maximum sustainable yield sets harvest levels in fisheries, whaling and forestry; k-selected species that breed late and slowly need strict protection and recover slowly; culling may be justified where predators are absent, as with Scottish red deer without wolves, or where introduced species threaten natives, while the reintroduction of wolves to Yellowstone shows how restoring a predator changes vegetation through the food web. Monitoring population size, distribution, survival and age structure measures success, and knowledge of adaptation and ranges of tolerance guides how abiotic conditions are managed. | |
| | The limits are uncertainty: natural fluctuations, unknown carrying capacities and the difficulty of predicting how communities respond. Ecological understanding cannot remove that uncertainty, but without it conservation decisions - to manage or not, to cull or protect, how much to harvest - are guesses. | |
| | **Weaker answer - Level 2 (5 marks)** | |
| | Succession is when a habitat changes over time from pioneer species to a climax community. Conservationists need to understand this because some habitats like heathland will turn into woodland if they are not managed by grazing or burning. Population dynamics is about how populations grow and are limited by factors like food and predators. This helps conservation because you can work out how many animals can be taken without the population declining, and when a population needs protecting. Monitoring populations shows if conservation is working. | |
| | **What makes the difference** | |
| | • Level 2 explains both concepts correctly and links them to decisions, but generally; Level 3 uses the technical vocabulary (seres, plagioclimax, k-selection, MSY) with named cases (Yellowstone wolves, Scottish deer, chalk grassland). | |
| | • Level 3 covers more applications (habitat creation, culling introduced species, adaptation and tolerance) and evaluates the uncertainties. | |
| | • Structure: Level 3 gives each concept its own paragraph with the decisions it informs, then a conclusion. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **10.2** | 1 mark for the point below.<br>**1** Any taxonomic group of organisms classified together by similarity (eg species, genus, family)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-85; AQA spec 3.1.3. | |
| **11.1** | **Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.**<br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br>**Indicative content - students are not expected to cover all of these**<br>• **Importance:** resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation)<br>• **Causes:** agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY<br>– trends by region<br>• **Consequences:** loss of resources, biodiversity and species, fragmentation<br>– hydrological changes<br>– soil impacts<br>– climate impacts (albedo, carbon, rainfall)<br>• **Productivity vs biodiversity:** non-indigenous species, monocultures, close planting, simple age structure, clear-felling<br>• **Sustainable management:** MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China)<br>– Rio Bravo case study | 25 |

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| Question | Marking guidance | Marks |
|---|---|---|
| | • **Evaluation:** economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures<br>– judgement<br>*Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*<br>**Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4. | |
| **11.2** | **Discuss how human activities influence biodiversity, using examples from a range of different contexts.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• **Direct exploitation:** food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil)<br>– deliberate eradication of predators, competitors, vectors, pests<br>• **Changes in abiotic factors:** water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage<br>– inactivity (stopping plagioclimax management)<br>• **Changes in biotic factors:** introduced competitors, predators, pathogens, hybridisation<br>– loss of pollinators, seed dispersers, keystone species<br>– food chain impacts (sea otters, puffins, jellyfish)<br>• **Habitat destruction:** deforestation, farmland expansion, urbanisation, mining, reservoirs<br>– fragmentation and edge effects<br>• **Contexts:** temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica | 25 |

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| Question | Marking guidance | Marks |
|---|---|---|
| | – agriculture, fishing, forestry<br>• **Positive influences:** habitat creation, conservation management, reintroductions, legislation<br>• Evaluation of which influences are most damaging and hardest to reverse<br>– conclusion<br>*Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*<br>**Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3. | |

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